Supplementary Note: Popcount as Hamming Weight and an Entropy-Oriented Diagnostic

Simon Tian

Abstract

This review-only supplement reframes the popcount-based Gaussianization argument in information-theoretic language. The main point is that popcount, more formally the Hamming weight, is an entropy-related diagnostic of first-order bit balance rather than a complete standalone quality measure for a random number generator. Matching the ideal Hamming-weight law is therefore useful evidence, but it does not by itself rule out dependence, structural artifacts, or predictability in the underlying source.

1 Review Status

This note is intended as a standalone supplement for internal review before any material is merged into the main manuscript. It leaves the main paper unchanged and instead collects one alternative interpretation of the current popcount analysis.

2 Popcount as Hamming Weight

Let $X = (X_1, \ldots, X_n) \in \{0,1\}^n$ be an n-bit output and define its Hamming weight

$$W_n = \sum_{i=1}^{n} X_i.$$

In the manuscript, W_n is the popcount of a VM-native output. Under an ideal unbiased independent-bit model, $X_i \sim \text{Bernoulli}(1/2)$ and

$$W_n \sim \text{Binomial}(n, 1/2).$$

Under the broader heterogeneous-bit model already used in the main paper, W_n is Poisson-binomial instead [2]. The central-limit-theorem argument in the manuscript then explains why the standardized Hamming weight is close to Gaussian under mild regularity conditions [1].

This perspective is useful because it isolates a specific question about the generator: do the bits appear balanced enough that the empirical weight law looks like the law predicted by fair coin flips?

3 An Entropy Decomposition

The information-theoretic interpretation comes from the fact that the Hamming weight identifies only the *composition* of the bit string, not the arrangement of the 1's inside it. For a fixed weight

w, the number of binary strings having exactly w ones is

$$\binom{n}{w}.$$

Conditioned on $W_n = w$, the remaining uncertainty is the choice of which w positions carry the ones, so

$$H(X \mid W_n = w) = \log_2 \binom{n}{w}.$$

This yields the standard entropy decomposition

$$H(X) = H(W_n) + H(X \mid W_n).$$

The consequence is immediate: the Hamming-weight variable compresses the original n-bit output very aggressively. For unbiased independent bits, $H(X) = n$ bits, whereas $H(W_n)$ grows only on the order of $\log n$. At the digest width used throughout much of the paper,

$$H(W_{256}) \approx 5.05 \text{ bits},$$

while the full 256-bit output has 256 bits of Shannon entropy under the ideal model.

So the Hamming-weight distribution is entropy-related, but only in a narrow sense: it measures whether the aggregate bit-balance behavior is compatible with high-entropy fair-coin outputs. It does *not* retain the full information content of the underlying string.

4 What This Means for RNG Quality

This provides an alternative interpretation of the current manuscript figures. The popcount diagnostics test whether the generator looks correct at a first-order level:

- are the bits centered near a fair split between zeros and ones?
- do the empirical weight probabilities follow the expected binomial/Poisson-binomial shape?
- do the quantiles and tails materially deviate from the ideal reference?

Those are meaningful quality checks, but they are not exhaustive. A generator can have a convincing Hamming-weight distribution and still be deficient in other ways: short-period structure, lag dependence, repeated patterns, or cryptographic predictability may still remain. That is why the Hamming-weight view is best read as a *first-order marginal diagnostic*, not as a full randomness certification. This is consistent with the broader philosophy behind local statistical batteries such as NIST SP 800-22, where useful statistical tests are treated as diagnostics rather than complete certification by themselves [3].

For that reason, an entropy-oriented framing should strengthen rather than replace the current evaluation logic. A practical interpretation is:

1. the Hamming-weight law checks bit balance and aggregate distributional fit;
2. dependence diagnostics such as repeated-sequence ACF remain necessary, because they probe structure not visible after reducing each output to a single weight;

3. if a scalar entropy-flavored score is desired, a cleaner candidate is the Kullback–Leibler divergence between the empirical weight law and the ideal reference law,

$$D_{\mathrm{KL}}(\widehat{P}_{W_n} \,\|\, P^{\star}_{W_n}) = \sum_{w} \widehat{P}_{W_n}(w) \log \frac{\widehat{P}_{W_n}(w)}{P^{\star}_{W_n}(w)},$$

where $P^{\star}_{W_n}$ is the ideal binomial or Poisson-binomial reference.

5 Connection to the Current Origin Figures

Figures 1–4 show how the entropy-oriented reading maps onto the refreshed Origin exports. The first three figures diagnose the Hamming-weight law directly, while the final figure shows why a separate dependence check is still needed.

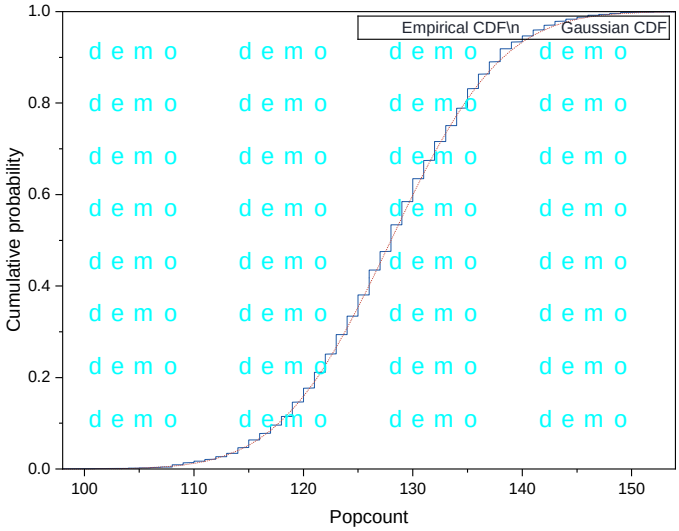


Figure 1: Empirical Hamming-weight CDF against the Gaussian/binomial-style reference. In the entropy-oriented reading, this figure checks whether the aggregate weight law is centered and shaped as expected under near-fair bit balance.

6 Suggested Positioning

If any of this is merged into the main manuscript, the cleanest claim is the following:

The Hamming-weight distribution is an entropy-related indicator of marginal bit balance, but it is not by itself a complete quality measure for the underlying random number generator.

That wording preserves the value of the current popcount analysis while making its scope explicit.

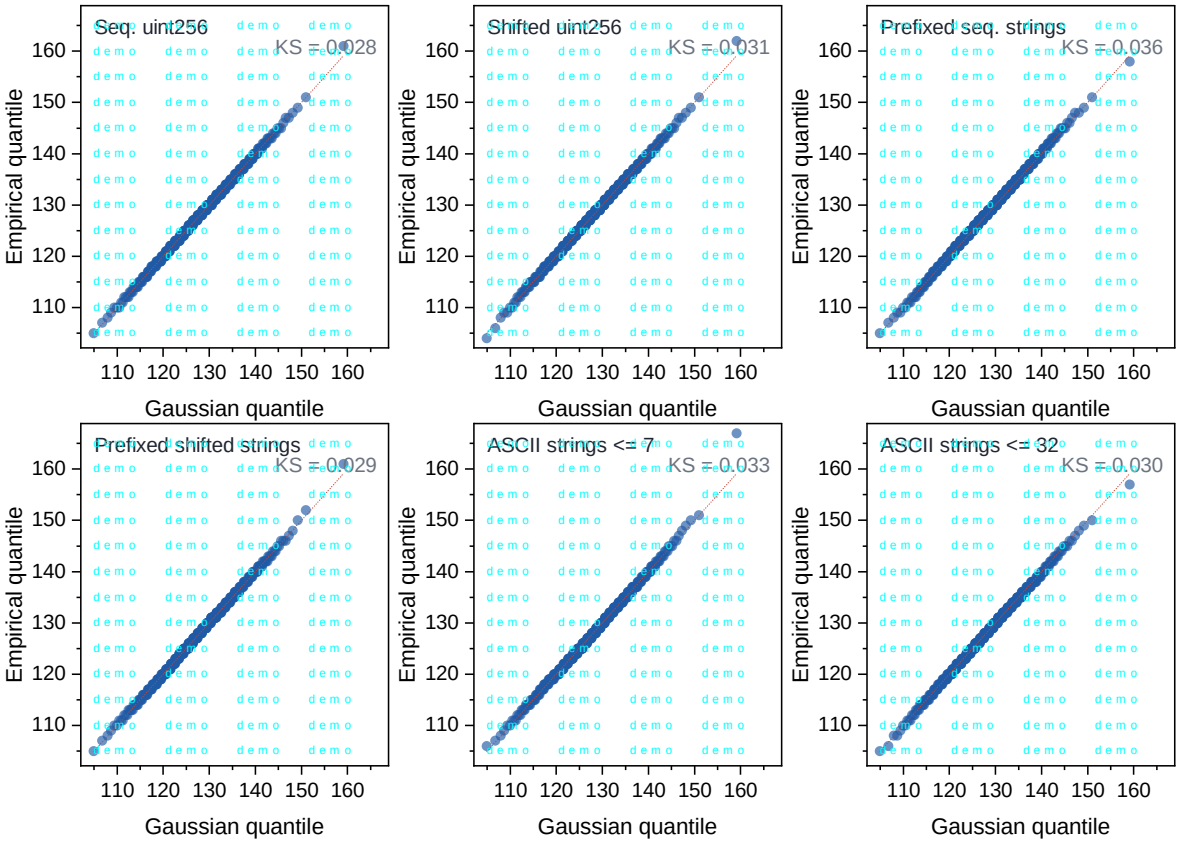


Figure 2: Hamming-weight quantile alignment across experiments. These panels are best read as a check on whether the observed type-class mass accumulates in the same way as the ideal reference law.

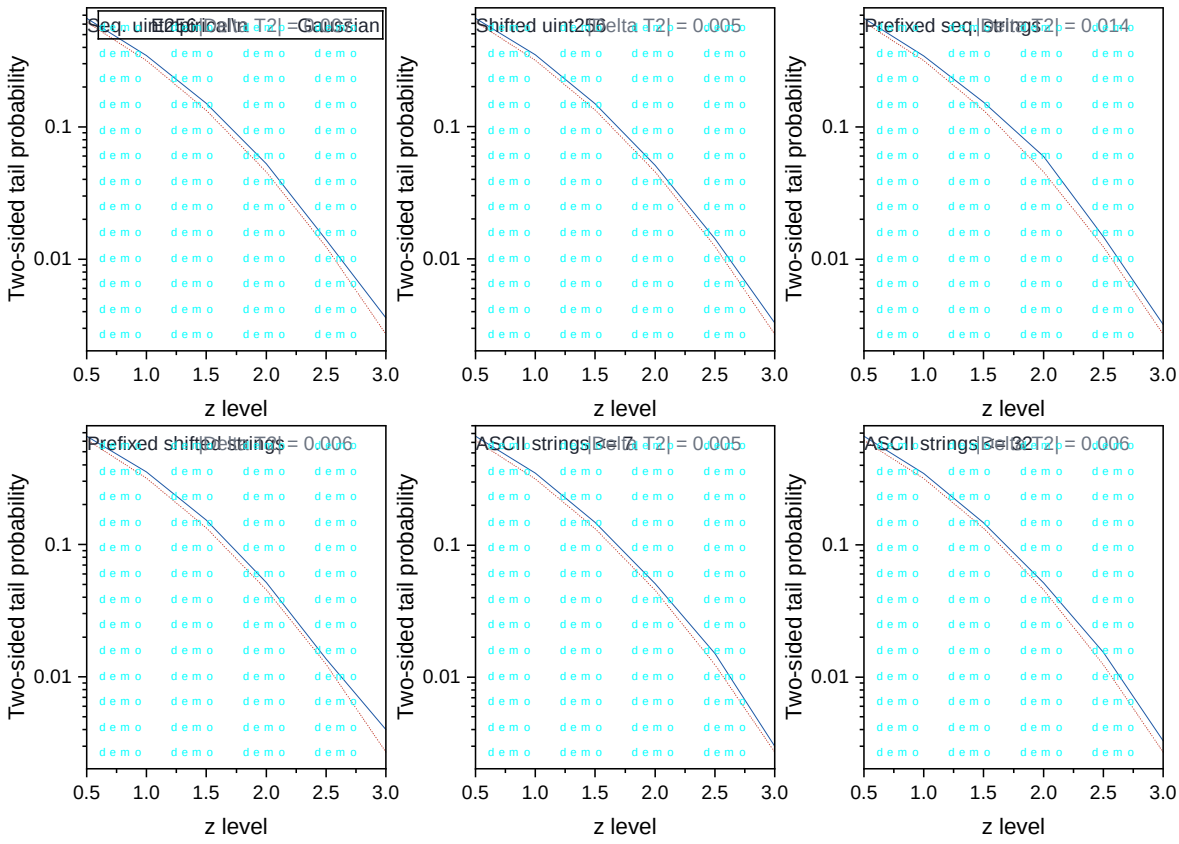


Figure 3: Hamming-weight tail probabilities across experiments. Tail behavior is important because it detects whether the generator allocates too much or too little probability to unusually light or unusually heavy outputs.

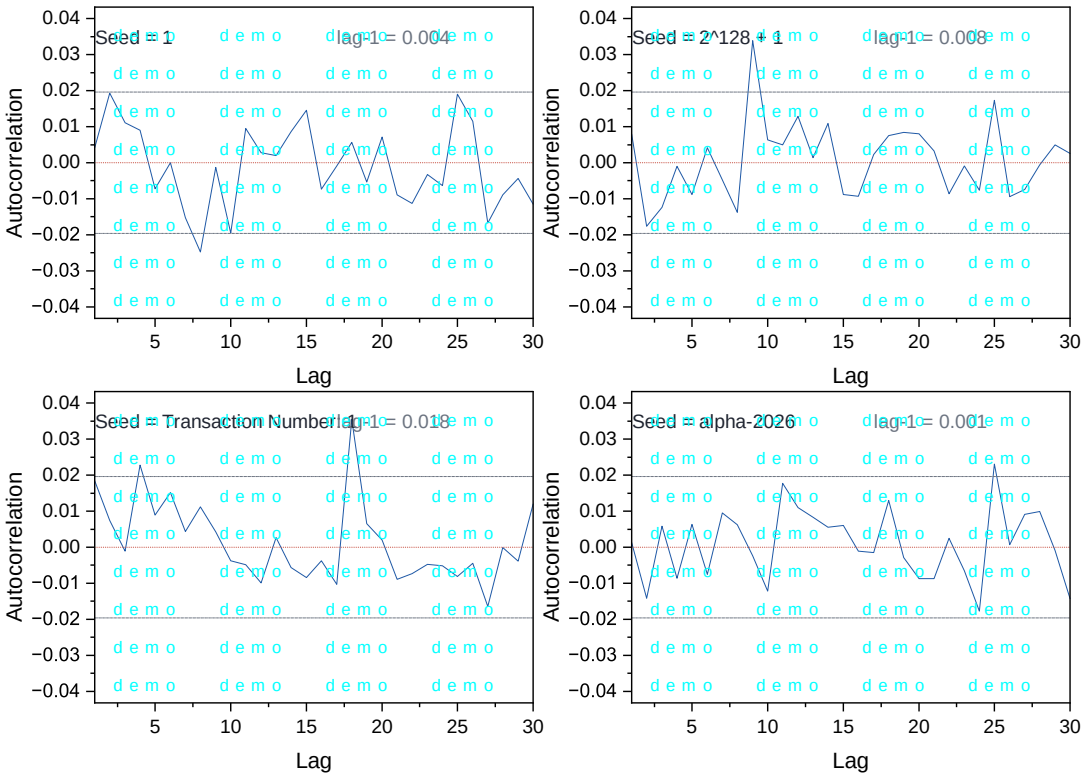


Figure 4: Repeated-sequence autocorrelation panels. This figure is not a Hamming-weight diagnostic; it is included to emphasize that good weight-law fit does not by itself rule out lag dependence or sequence structure.

References

[1] Rick Durrett. *Probability: Theory and Examples*. Cambridge University Press, 5 edition, 2019. doi: 10.1017/9781108591034.

[2] Yili Hong. On computing the distribution function for the poisson binomial distribution. *Computational Statistics & Data Analysis*, 59:41–51, 2013. doi: 10.1016/j.csda.2012.10.006.

[3] Andrew Rukhin, Juan Soto, James Nechvatal, Miles Smid, Elaine Barker, Stefan Leigh, Mark Levenson, Mark Vangel, David Banks, Nancy Heckert, James Dray, San Vo, and Lawrence Bassham. A statistical test suite for random and pseudorandom number generators for cryptographic applications. Special Publication 800-22 Rev. 1a, National Institute of Standards and Technology, 2010.